Bubble CPAP Liquid Level Monitoring Device

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DUKE ENGINEERING DESIGN POD

Introduction

Background:

BCPAP, or *bubble continuous positive airway pressure*, is a noninvasive respiratory support device that supports the lung health of babies by providing humidified air. The BCPAPs used in Ugandan hospitals are low cost and low maintenance making them crucial devices to treat babies in underdeveloped countries. BCPAP devices contributed to a 21% absolute improvement in the survival of babies needing respiratory support. Consequently, having any of these devices run without their needed inputs of water can greatly impact the health of newborns in Ugandan NICUs.

Problem Statement:

Due to a lack of staff in Ugandan hospitals, nurses are often too busy to consistently check on the water level of BCPAP machines. Our goal is to design a system that alerts nurses when the liquid level in BCPAP devices is critically low.

Design Blocks and Materials

Attachment Mechanism	Alert System	Sensor System
- Purpose: **Physical integration** of the sensor and alert systems with the BCPAP cylinder - Implementation: Stable **wood** platform with the **circuit secured and enclosed within**	- Purpose: Uses **data from sensor** to **notify nurses** when liquid level is **below tared threshold** - Implementation: **3 LEDs** and **vibration buzzer**	- Purpose: **Detects** the **amount of water** within BCPAP device - Implementation: **Load cell weight sensor** with **Arduino Nano**

Final Design Solution

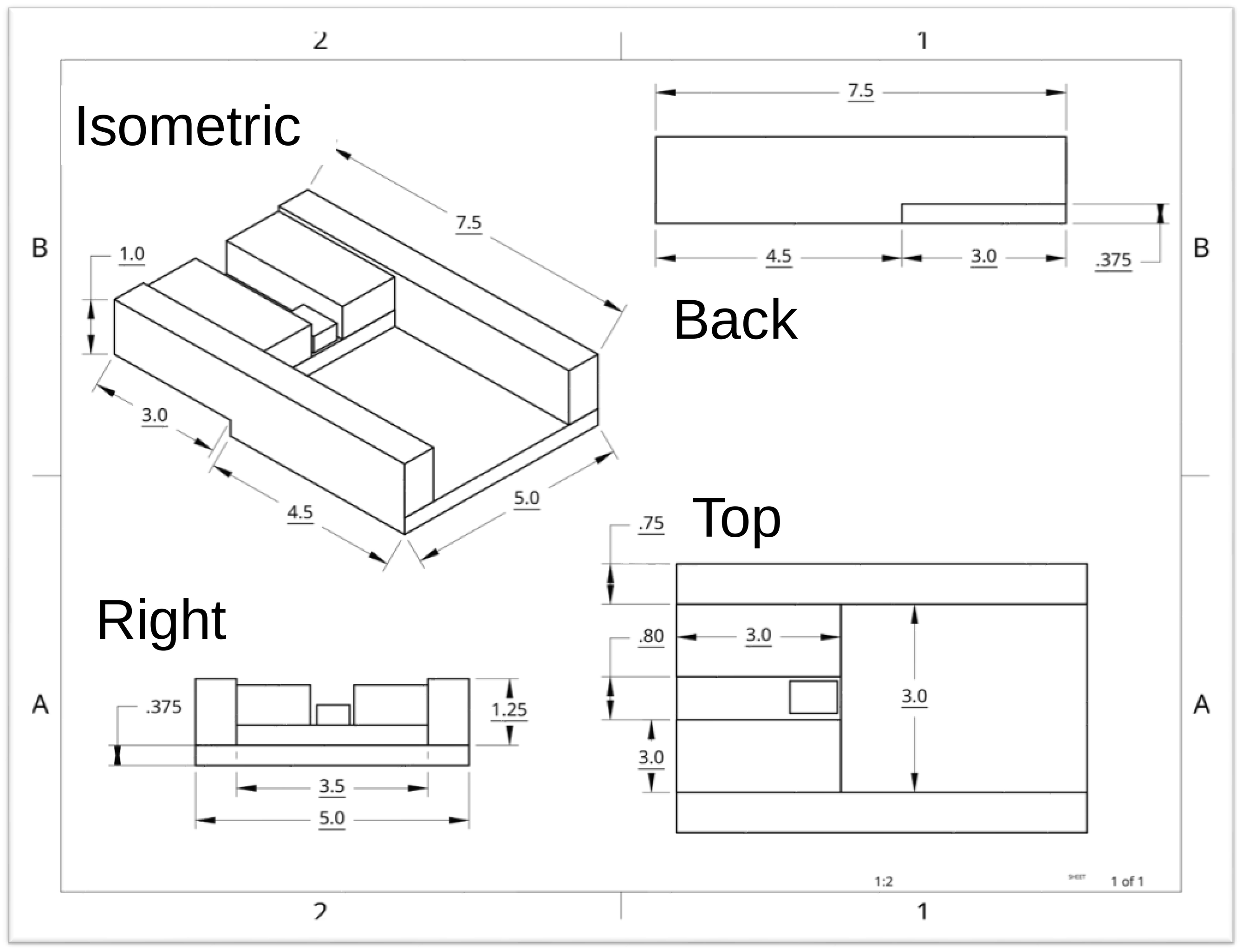

Figure 1: Dimensions and Different Views of Final Wood Stand for BCPAP modeled using CAD

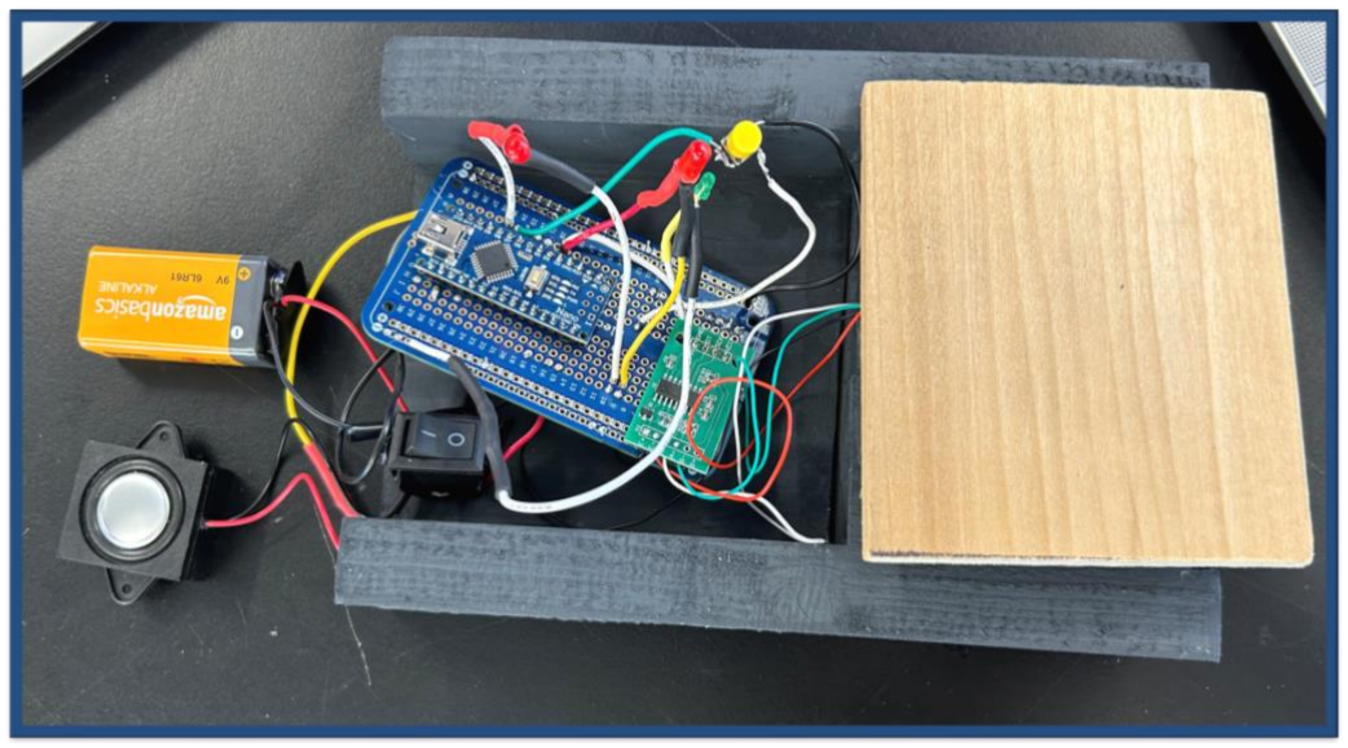

Figure 2: Dimensions and Different Views of Final Wood Stand for BCPAP modeled using CAD

Earlier Design Prototypes

Figure 3: First Iteration 3D Printed Enclosure Stand for BCPAP Device

Figure 4: Second Iteration of 3D Printed Enclosure Stand

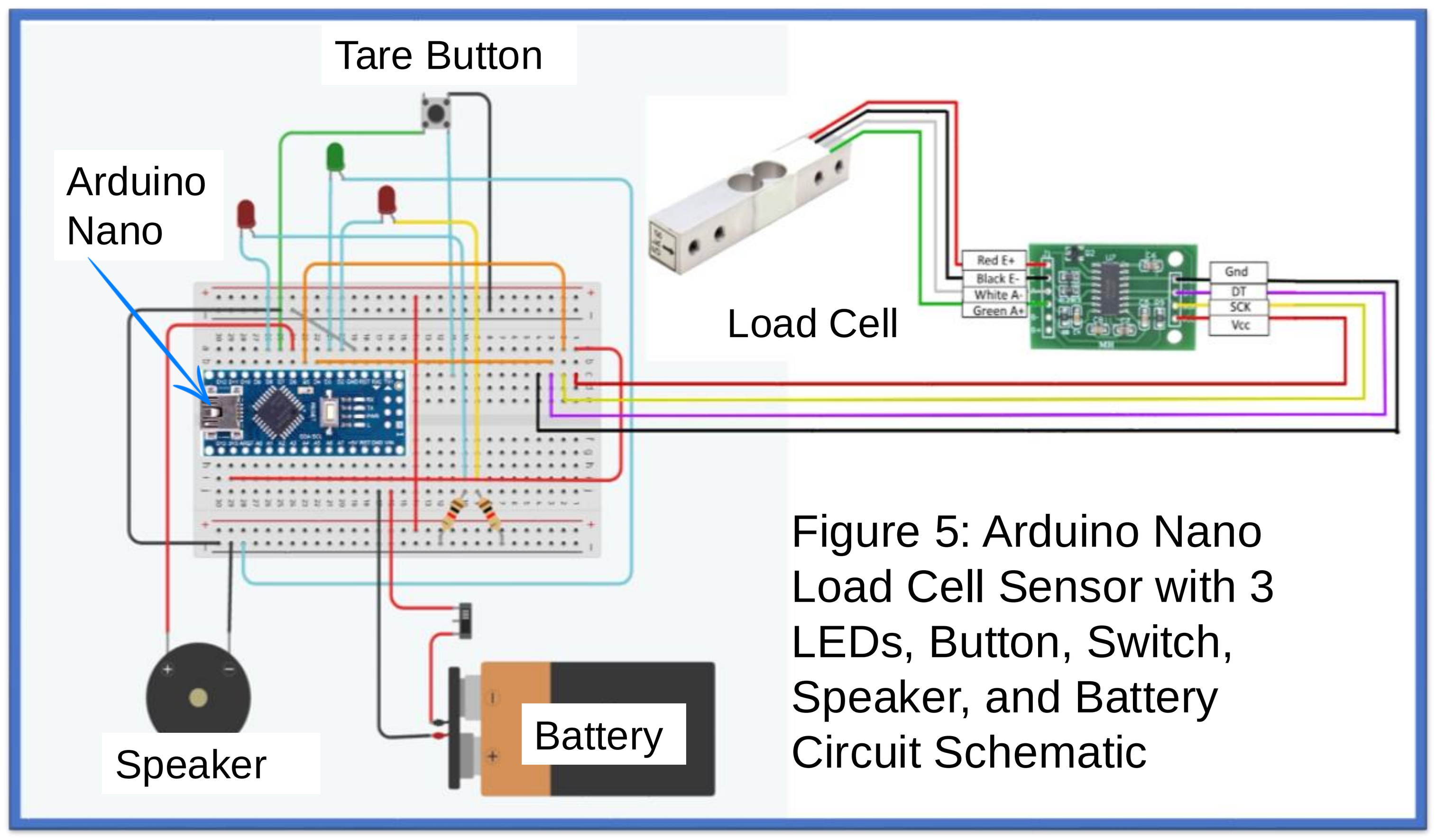

Figure 5: Arduino Nano Load Cell Sensor with 3 LEDs, Button, Switch, Speaker, and Battery Circuit Schematic

Design Criteria and Testing

Design Criteria	Target value	Test	Result
Locally Manufacturable	100% of materials can be sourced in Uganda	Evaluate the design components; the test will run on a pass/fail scale	Pass
Cost	Less than $10	Evaluate the cost of design components; The test will run on a pass/fail scale based of the target value of $10	No Pass - Total Cost: 65000 Ugandan shillings or $17.58 US Dollars
Durability	Can be used for 500 BCPAP refills before a need for replacement	Add weight, tare, and remove weight 500 times	TBD
Ease of Use	Nurses can understand and maintain device with minimal support from engineers	Subjects will assess, on a user defined scale, how easy the device is to set up and remove	Pass
Portability	<1 minutes to remove device <1 minute to attach device	Subjects are timed setting up and removing our device	Pass
Small Size	1' X 1' X 1' or smaller in size	Size is measured by our team	Pass
Aesthetics	>85% of people polled confirm that the device looks reliable/trustworthy	Subjects rate aesthetics using a likert scale.	Pass

Conclusion

Our design passed most of the tests we conducted. This indicates our product's success as a proof-of-concept for a BCPAP water level monitoring device.

Moving forward, we hope to:

- **Create** a **cheaper** alternative to Arduino
- **Develop** easier-to-manufacture stands
- **Enhance** the system's **durability and usability** in hospital environments
- **Conduct testing** in Uganda to gather real-world feedback
- **Explore partnerships** for scalable production and distribution in Uganda

This solution addresses the urgent need for cost-effective and efficient water level monitoring systems in the low-resource Ugandan NICUs, improving neonatal care.

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